Day 22 02-03-2026 Monday.

Selection Sort.

Assumption (small) min

9 2 6 16 13 → 2 6 9 13 16 → Final Answer

small

compare

Sort → Ascending order

Selection Sort

Sort → Ascending order

→ Something you will select

↳ you are going to select a number → Minimum

① Assume 1st element a minimum

② Compare minimum_Assumed with next element

↳ Big (or) small

9 > 2 ✓

min > next

Selected > next no, yes

Swap

9 2

---

compare

2 9 6

Smallest min

9 > 6 ✓ Selected > next

min > next yes

Swap

---

compare

2 6 9 16

min

9 > 16 X Selected > next

min > next No

No swap

Compare

2 6 9 | 16 13

min

16 > 13 ✓ Selected > next

min > next yes

Swap

Final Answer | 2 | 6 | 9 | 13 | 16 |

	9	2	6	16	13
index –	0	1	2	3	4

size = 5

read ↗ ↑ comparison

a[0] = 9

min = a[0]

for comparison → for loop

ಕೊನೆವರೆಗೂ 1 2 3 4

iteration ಗೆ → for loop

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0 1 2 3

read (no need to read last element)

```
for (int i = 0; i < size - 1; i++)
    for (int j = i+1; j < size; j++)
```

(Big) > Small → Sort

selected > next

Swap

Big Small → Sort → 2 9

9 > 2

Small Big.

9 > 2

English (Algorithm)

#1) Take a list/array

0				
9	2	6	16	13

#2) find size — size = 5 (helps in loop)

#3) Iterate through each element

min_value = array[0] ⇒ min_value = array[i]

min_index = 0; ⇒ min_index = i

#4) for comparing min_value with next we need 2nd loop that starts from 1st index (i+1) and goes till end.

9 2

i = 0 j = 1

check :- if (min_value > array[j]) → if yes

{

min_value = number[j];

min_index = j

}

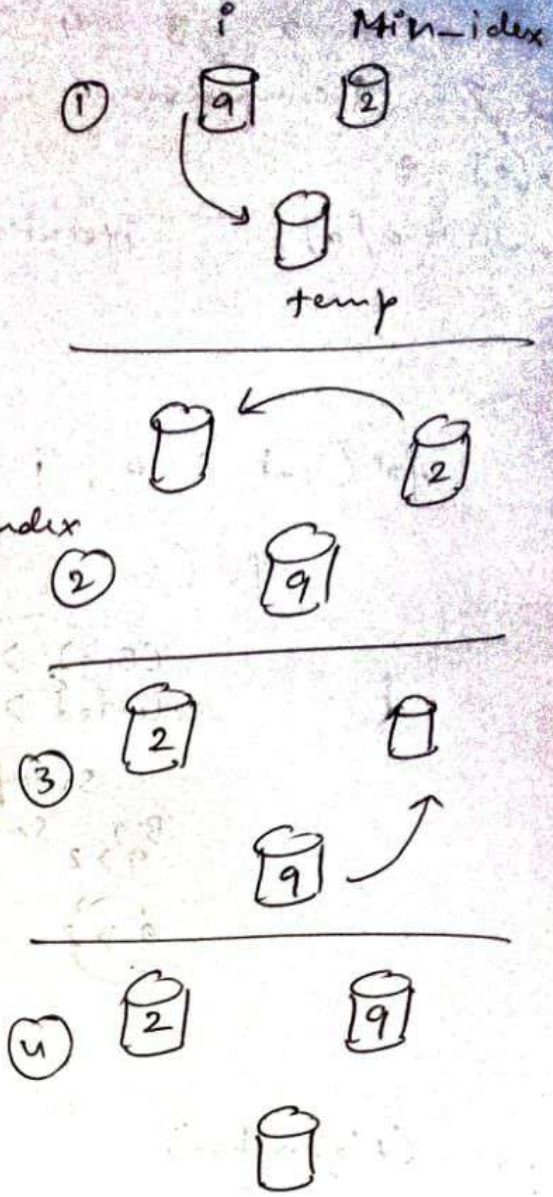
#5) swap.

~~temp = a[i]~~
~~a[i] = arr[i+1]~~
~~a[i+1] = a[~~

temp = array[i]
array[i] = array[~~i++~~]
array[~~i++~~] = temp.

min_index

min_index



```
Public class Selection_Sort {
  Public static void SelectionSort (int[] array) {
      for(int i=0 ; i<array.length-1 ; i++) {
            int min_val = array[i];
            int min_index = i ;
          for(int j= i+1 ; j<array.length ; j++) {
                if (min_val > array[j]) {
                      min_val = array[j] ;
                      min_index = j ;
                }
          }
          int temp = array[i];
          array[i] = array[min_index];
          array[min_index] = temp ;
      }
      for (int ~~j=i+1~~ i=0 ; i<arr.length ; i++) {
            System.out.print ( array[i] + " ");
      }
      System.out.println ();
  }
  Public static void main (String[] args) {
        int[] array = {9, 2, 6, 16, 13};
        SelectionSort (array);
  }
}
```